
Sheaf Neural Networks for E-Commerce Product Recommendation: Applying Sheaf4Rec to Amazon Reviews 2023

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Abstract

Most graph-based recommenders encode every user and product as a single embedding vector in one shared latent space. This project asks whether a richer representation helps: instead of one vector per node, we assign each user and product an entire vector space and learn linear maps that describe how those spaces relate along each interaction. This is the idea behind Sheaf4Rec, a recommender built on cellular sheaf theory and neural sheaf diffusion. We apply it to the All Beauty category of Amazon Reviews 2023, building a user-item bipartite graph from raw review records and treating top- K recommendation as link prediction. We will benchmark Sheaf4Rec against a popularity baseline, a standard GNN, and LightGCN, and study whether the sheaf representation buys us better ranking quality, more stable behaviour as the network deepens, and a favourable accuracy-efficiency trade-off on sparse real-world data.

1 Application Domain

The domain is e-commerce product recommendation, specifically beauty products. Shoppers leave ratings and reviews on items they buy, and a recommender’s job is to predict which unseen products a user would rate highly. We cast this as link prediction on a bipartite graph whose two node types are users and products and whose edges are observed interactions; the deliverable for each user is a ranked list of top- K products.

The reason this domain is interesting for a graph ML course is that a user’s taste in beauty products is rarely one-dimensional - the same person may buy skincare for sensitivity, makeup for occasion, and hair care for a specific texture. A single embedding vector has to average all of these signals into one point. We want to test whether letting each node carry a small vector space, rather than a single vector, captures that context better and yields recommendations that reflect distinct facets of a shopper’s behaviour.

2 Dataset, Task, and Metrics

Dataset. We use Amazon Reviews 2023 – All Beauty from HuggingFace. Each record links a user to a product through a rating, together with review text, a timestamp, and product metadata. Since the raw data is a table of reviews rather than a graph, we construct a bipartite interaction graph

$$G = (U, I, E), \quad (u, i) \in E \iff \text{user } u \text{ reviewed or rated item } i, \quad (1)$$

where U is the user set, I the item set, and E the interaction set. We reduce explicit ratings to implicit feedback,

$$R_{ui} = \begin{cases} 1, & \text{rating}_{ui} \geq 4, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

so that an edge signals a genuine positive interaction. To keep the graph learnable and reduce extreme sparsity we will apply k -core filtering (dropping users and items with very few interactions) and split each user’s interactions chronologically or randomly into train/validation/test.

Task. The task is top- K recommendation. The model scores every candidate item for a user and returns the K highest-scoring products:

$$\hat{y}_{ui} = f(u, i), \quad \text{TopK}(u) = \{i_1, i_2, \dots, i_K\}. \quad (3)$$

In Sheaf4Rec, f is realised by producing final user and item representations $\mathcal{F}^u$ and $\mathcal{F}^v$ and computing a score matrix $S = (\mathcal{F}^u)^\top \mathcal{F}^v$, then ranking each user’s row.

Metrics. We report Recall@ K , Precision@ K , NDCG@ K , and Hit Ratio@ K , with Recall@10 and NDCG@10 as the primary numbers because they measure whether relevant items are both retrieved and ranked highly. To match the evaluation used in the Sheaf4Rec paper we additionally compute F1-Score@ K and MRR@ K , and we log recommendation time so we can compare inference cost across models:

$$\text{Recall@}K = \frac{|\text{Rec@}K \cap \text{Rel}|}{|\text{Rel}|}, \quad \text{NDCG@}K = \frac{\text{DCG@}K}{\text{IDCG@}K}. \quad (4)$$

Why this dataset? All Beauty is a compact, self-contained slice of Amazon Reviews 2023 that still exhibits realistic e-commerce sparsity, so it is tractable on a course budget while remaining representative. It also has a practical advantage for us: the Sheaf4Rec paper itself evaluates on an Amazon Beauty dataset, which gives us a published reference point to sanity-check our reproduction against. Finally, the presence of ratings and timestamps lets us go beyond a plain binary graph and experiment with rating-aware or time-aware edge construction.

3 Graph ML Techniques and Models

We compare three tiers of model: a non-graph baseline, standard graph recommenders, and the sheaf-based main model. LightGCN — the current go-to graph recommender — is deliberately placed as a baseline so we can isolate what the sheaf structure adds.

Model	Type	Role
Popularity	Non-personalized	Simple baseline
LightGCN	Graph recommender	Strong baseline
Sheaf4Rec	Sheaf neural network	Main model

Table 1: Planned model comparison.

3.1 Main Model: Sheaf4Rec

The distinguishing move in Sheaf4Rec is to replace single-vector node embeddings with *stalks* — a vector space $\mathcal{F}(v)$ attached to each node and $\mathcal{F}(e)$ to each edge — drawn from cellular sheaf theory. In the recommender, vertex stalks (0-cochains) hold user embeddings and edge stalks (1-cochains) hold item embeddings, both initialised from trainable embedding tables Ψ^u and Ψ^v .

For every incident node–edge pair, a *learnable restriction map* $\mathcal{F}_{v \trianglelefteq e} : \mathcal{F}(v) \rightarrow \mathcal{F}(e)$ projects a node’s data into the edge’s space, encoding how a user’s latent preferences line up with an item’s latent semantics. Disagreement across an edge is measured by the co-boundary map,

$$\delta(x)_e = \mathcal{F}_{v \trianglelefteq e} x_v - \mathcal{F}_{u \trianglelefteq e} x_u, \quad (5)$$

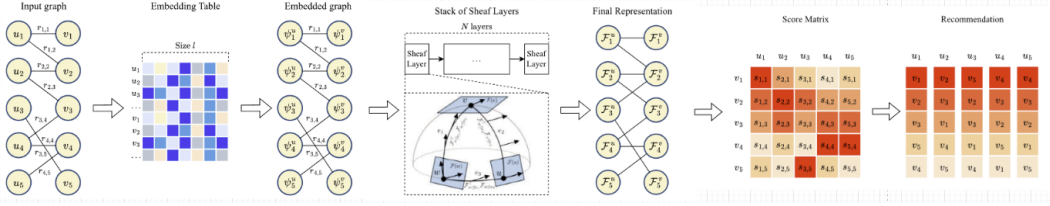


Figure 1: **The Sheaf4Rec pipeline.** First, the raw user–item interactions are turned into a bipartite graph and mapped to an embedded graph via two embedding tables. Next, the embedded graph is passed through a stack of sheaf layers that continuously update the user (0-cochain) and item (1-cochain) representations. Finally, the resulting representations are combined into a score matrix $S = (\mathcal{F}^u)^\top \mathcal{F}^v$, whose entries are ranked to produce a top- K recommendation list for each user. Figure reproduced from Purificato et al. [1].

and the (normalized) sheaf Laplacian generalises the ordinary graph Laplacian by routing connectivity through these maps rather than through raw edges:

$$L_{\mathcal{F}} = \delta^\top \delta, \quad \Delta_{\mathcal{F}} = D^{-1/2} L_{\mathcal{F}} D^{-1/2}. \quad (6)$$

Stacking N sheaf layers runs a neural sheaf diffusion process in which the learned sheaf — and therefore the effective geometry of the graph — adapts at every layer:

$$X(t+1) = X(t) - \sigma(\Delta_{\mathcal{F}}(t) (I \otimes W_1) X_t W_2). \quad (7)$$

Because a strictly bipartite graph is not directly compatible with the sheaf Laplacian, we follow the paper’s projection that preserves sheaf properties while keeping parameters trainable. Training uses Bayesian Personalized Ranking over observed edges (positives) and sampled unobserved edges (negatives):

$$\mathcal{L}_{\text{BPR}} = - \sum_{(u,i,j)} \ln \sigma(\hat{y}_{ui} - \hat{y}_{uj}) + \lambda \|\Theta\|^2, \quad (8)$$

with i a positive and j a negative item for user u . We will start from the paper’s vanilla sheaf-diffusion configuration and tune stalk dimension and layer count.

4 Why Sheaf4Rec Is Appropriate

Amazon review data gives us little beyond user IDs, item IDs, and interactions, there are no rich, reliable node features. This is exactly the collaborative-filtering regime where graph recommenders shine, and where Sheaf4Rec’s contribution is most visible, because it improves the *representation* itself rather than depending on external features. Assigning each node a vector space plus learned restriction maps lets the model express context-dependent preference, different facets of a user surfacing in different neighbourhoods, which a single shared embedding cannot.

The choice is also well motivated relative to our baselines. LightGCN and NGCF all collapse every node into one point in a global space and are known to over-smooth as layers are added, so they are typically kept shallow. Sheaf diffusion is specifically designed to resist over-smoothing, which lets Sheaf4Rec exploit deeper stacking (the paper finds its best results around five layers) and therefore reach information further along multi-hop User $\rightarrow$ Product $\rightarrow$ User $\rightarrow$ Product paths. On the deployment side, the sheaf parameterisation makes training somewhat heavier but inference lighter, since recommendation uses pre-optimised local maps instead of recursive message passing, which is a sensible trade-off for e-commerce, where models can be trained offline but must serve fast. Keeping LightGCN as a baseline rather than the headline model means any gain we observe can be attributed to the sheaf representation on identical data and splits.

5 Planned Analysis

Our analysis targets both mechanism and outcome. On outcomes, we will test whether Sheaf4Rec beats popularity, LightGCN, and a standard GNN on Recall@10, NDCG@10, F1@ K , and MRR@ K ,

and whether it holds a better precision–recall balance rather than winning one at the cost of the other. On mechanism, we will vary the number of sheaf layers to check the claimed robustness to over-smoothing, and sweep the stalk/latent dimension to map the accuracy-versus-cost curve. We plan a small expressiveness study using the paper’s framing, configuring stalk sizes so the model reduces to a GCN-like $(N, 1)$ or GAT-like $(1, N)$ form to confirm that the full sheaf (N, N) is what drives any improvement. We will measure training and inference time to verify the “slower to train, faster to serve” trade-off on our data. Finally, because Amazon Reviews carries ratings and timestamps, we will test whether rating-aware or time-aware edge construction improves recommendation quality over the plain binary graph, which was a direction that the original paper’s datasets exploit less, and a natural place for our project to add something of its own.

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